

Ethics for Engineers (with emphasis in Computer Science)

Javier Larrosa

UPC

larrosa@cs.upc.edu

April 16, 2026

Societal Impact of Technology

Technological development is having a huge impact in society

Tech Companies

Technological progress has to be done **legally** and **ethically**. For that purpose companies are hiring **lawyers** and **ethicists**

Motivation

The motivation for lawyers is obvious, but what is the real motivation for ethicists? (ethics washing?)

Silicon Valley's attitude

It is better to ask for forgiveness than to ask for permission

Silicon Valley's attitude

Move fast and break things (namely, it's more worthwhile to make mistakes and disrupt technologies along the way than to play it safe at a slow and steady pace.)

The New York Times

The Times Sues OpenAI and Microsoft Over A.I. Use of Copyrighted Work

Millions of articles from The New York Times were used to train chatbots that now compete with it, the lawsuit said.



Share full article



1.3K



THE WALL STREET JOURNAL

English Edition | Print Edition | Video | Audio | Latest Headlines | Puzzles | More

Business | U.S. | Politics | Economy | **Tech** | Markets & Finance | Opinion | Free Expression | Arts | Lifestyle | Real Estate

TECHNOLOGY

Meta and YouTube Lose Landmark Social-Media Addiction Trial

Jurors found the companies negligent and said their app designs caused harm to children

By [Meghan Bobrowsky](#) [Follow](#), [Erich Schwartzel](#) [Follow](#) and [Katherine Sayre](#) [Follow](#)

Updated March 25, 2026 4:15 pm ET



Share



Resize



538

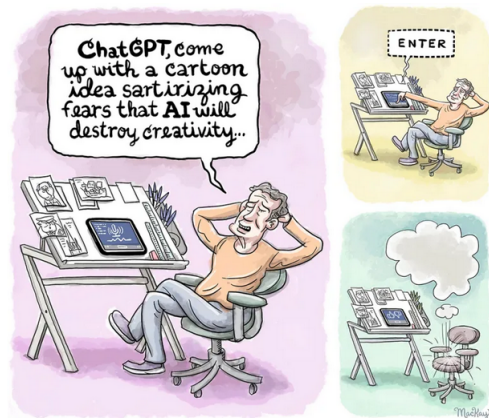


Listen (2 min)



Motivation

April 14, 2023 by Graeme MacKay



Editorial Cartoon by Graeme MacKay, For the Globe & Mail – Saturday April 15, 2023

Should Comp. Scientists Care About Ethics?

ACM's president, The Conversation 2018

Technological professionals are the first, and last, lines of defense against the misuse of technology. Nobody else **understands** the systems as well, and nobody else is in a position to **protect** specific data elements or ensure the connections between one component and another are **appropriate, safe and reliable**. As the role of computing continues its decades-long expansion in society, **computer scientists are central to what happens next.**

- Regarding Ethical aspects we do not have **the** responsibility, but we have **some** responsibility.
- We need some skills in ethics to participate in the discussion

- The Responsibility of Engineers.
 - Responsibility (Blameworthiness)
 - Engineers vs Managers
 - Whistle-blowing
 - Codes of Conduct
- Humane Decision making in Innovation
 - The Ethical Cycle
 - The Three Laws of Humane Tech
 - Ethical Questions in Design
- Artificial Intelligence
 - EU AI Act (2024)
 - AI risks

The Responsibility of Engineers



The Responsibility of Engineers

Boeing 737 MAX crash



Progress has risks, should we stop progress?

- It is unreasonable to believe that innovation has zero-risk.
- From a utilitarian perspective, a small risk is clearly outweighed by the long-term benefits
- From a deontological perspective, a small risk may be accepted as long as decisions have been made thoughtfully and risks have been minimized and are very small.

Progress has risks, should we stop progress?

- It is unreasonable to believe that innovation has zero-risk.
- From a utilitarian perspective, a small risk is clearly outweighed by the long-term benefits
- From a deontological perspective, a small risk may be accepted as long as decisions have been made thoughtfully and risks have been minimized and are very small.

Liability

Even though we can accept that progress has risks, if something goes wrong, it is imperative to know if there is somebody to be blamed.

Definition (The problem of Many hands)

When many people are involved in an activity, it is often difficult, if not impossible, to pinpoint who is morally responsible for what,

When work is done by teams, it should be clear who is responsible for what

- To guarantee that nothing is left unattended.
- To avoid the problem of many hands

Responsible Engineers

As professionals we want to guarantee that nothing goes wrong and, if it does, we want to feel that our behaviour was correct.

- **Passive** Responsibility: for harm that already happened
 - This is about accountability (blameability)
- **Active** Responsibility: for harm that may happen in the future
 - This is about how to behave

Passive Responsibility

Whenever something goes wrong the question who is (legally and/or morally) responsible for it often quickly arises

Passive Responsibility

Whenever something goes wrong the question who is (legally and/or morally) responsible for it often quickly arises

Usually **four conditions need to apply** for an actor to be held responsible:

Passive Responsibility

Whenever something goes wrong the question who is (legally and/or morally) responsible for it often quickly arises

Usually **four conditions need to apply** for an actor to be held responsible:

- **Wrong-doing:** a moral norm has been violated or a value has been threaten (remember ethical theories)

Passive Responsibility

Whenever something goes wrong the question who is (legally and/or morally) responsible for it often quickly arises

Usually **four conditions need to apply** for an actor to be held responsible:

- **Wrong-doing:** a moral norm has been violated or a value has been threaten (remember ethical theories)
- **Causal Contribution:** the consequences of the action (or inaction) must be reasonably related to the harm

Passive Responsibility

Whenever something goes wrong the question who is (legally and/or morally) responsible for it often quickly arises

Usually **four conditions need to apply** for an actor to be held responsible:

- **Wrong-doing:** a moral norm has been violated or a value has been threaten (remember ethical theories)
- **Causal Contribution:** the consequences of the action (or inaction) must be reasonably related to the harm
- **Foreseeability:** the harmful consequences of the action should be reasonably predictable

Passive Responsibility

Whenever something goes wrong the question who is (legally and/or morally) responsible for it often quickly arises

Usually **four conditions need to apply** for an actor to be held responsible:

- **Wrong-doing:** a moral norm has been violated or a value has been threaten (remember ethical theories)
- **Causal Contribution:** the consequences of the action (or inaction) must be reasonably related to the harm
- **Foreseeability:** the harmful consequences of the action should be reasonably predictable
- **Freedom of action:** an alternative course of action was available

Example

At a party

John was in a party. His friend Peter, who was drunk, drove home. On the way, he lost the car control and had an accident.

Example

At a party

John was in a party. His friend Peter, who was drunk, drove home. On the way, he lost the car control and had an accident.

The accident happened, so moral analysis is about **passive** responsibility

Example

At a party

John was in a party. His friend Peter, who was drunk, drove home. On the way, he lost the car control and had an accident.

The accident happened, so moral analysis is about **passive** responsibility

- Is **Peter** to be blamed?

Example

At a party

John was in a party. His friend Peter, who was drunk, drove home. On the way, he lost the car control and had an accident.

The accident happened, so moral analysis is about **passive** responsibility

- Is **Peter** to be blamed?
- Is **John** to be blamed?

Example

At a party

John was in a party. His friend Peter, who was drunk, drove home. On the way, he had an accident.

What if,

- Peter was not drunk (foreeability)
- John was not at the party, but just a bystander on the street (wrong-doing)
- John did not see Peter leaving (freedom of action)
- Peter's drunkenness did not have any effect in the accident (causal contribution)

Example

Make a moral Judgment of the following story

To become a police officer, people need to pass several exams about a variety of topics. Maria is a police officer in Barcelona, but her Catalan is very poor and she paid somebody to impersonate her in her Catalan exam during the admission process. Some months after becoming an officer she was involved in a shooting and some innocent bystander was shot and injured. She blames herself for that, because she says that she is a fraudulent police.

Example

Solution

Cheating in the exam is doubtlessly morally wrong (also illegal) and this is easy to justify with any ethical theory. However, if we analyze if **Maria is blameable for the shooting incident** we need to see to which extend the following conditions hold (with the information that we have): 1.

Wrong-doing: the only wrong thing that Maria did is cheating in the exam. Yes, this is wrong. 2. **Causal contribution:** not having a high level of Catalan does not seem to have any effect on how things went in the shooting incident, as far as we know. (It would be different had Maria cheated in other exams such as a physical test or a shooting test) 3.

Forseability: it is difficult to justify that cheating in a Catalan exam may end up causing harm in a shooting. 4. **Freedom of action:** Maria was not forced to cheat, so she clearly could have chosen not to do it. In summary, Maria voluntarily did something wrong, but it is unconnected with the shooting incident, so **she is not to be blamed for it**

Active Responsibility

Act taking into account that you do not want to be blamed if anything goes wrong. That is,

Active Responsibility

Act taking into account that you do not want to be blamed if anything goes wrong. That is,
(If you are free to chose) Do not break any (deontological) rule, such that breaking it (is likely to) causes harm

Active Responsibility

Act taking into account that you do not want to be blamed if anything goes wrong. That is,
(If you are free to chose) Do not break any (deontological) rule, such that breaking it (is likely to) causes harm

Example: in the two previous examples, active responsibility's would be to identify if there is something ethically wrong in the story even if the bad thing did not happen.

Would John be blameable even if Peter did not have an accident?

Would Maria be blameable even if nobody was shot?

Active Responsibility: Engineers vs Managers

- Engineers are usually hierarchically below managers and sometimes they disagree.
- How to deal with (potential) ethical conflicts?:

Active Responsibility: Engineers vs Managers

- Engineers are usually hierarchically below managers and sometimes they disagree.
- How to deal with (potential) ethical conflicts?:
 - **Separatism**: Engineers are responsible for the design and development and not for wider social consequences (critique: conflict with slide 5)

Active Responsibility: Engineers vs Managers

- Engineers are usually hierarchically below managers and sometimes they disagree.
- How to deal with (potential) ethical conflicts?:
 - **Separatism**: Engineers are responsible for the design and development and not for wider social consequences (critique: conflict with slide 5)
 - **Technocracy**: Let engineers decide (critique: engineers may miss interesting alternative points of view, or may be bias by their tech enthusiasm)

Active Responsibility: Engineers vs Managers

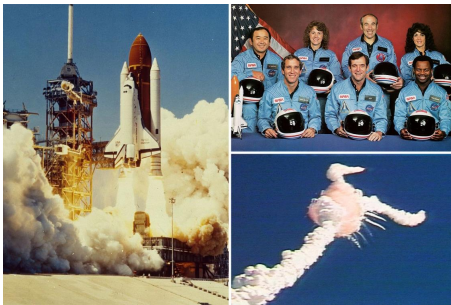
- Engineers are usually hierarchically below managers and sometimes they disagree.
- How to deal with (potential) ethical conflicts?:
 - **Separatism**: Engineers are responsible for the design and development and not for wider social consequences (critique: conflict with slide 5)
 - **Technocracy**: Let engineers decide (critique: engineers may miss interesting alternative points of view, or may be biased by their tech enthusiasm)

The previous approaches are not satisfactory. The preferred solution is,

Cooperation

A Company should have **dialog channels** where Engineers and Managers exchange their perspectives

(Lack of) Dialog Channels: example



1986 Challenger's Disaster

Famous Physicist Richard Feynman was one of the members of the investigation committee.

Feynman's account reveals a **disconnect** between NASA's **engineers** and **executives** that was far more striking than he expected. Sometimes they even used the same word with different meanings.

Engineers vs Managers: Example

Sigal Samuel's blog *Future Perfect*, Reader question

I am an engineer working at a medium-sized ad agency, mostly on non-generative machine learning models. Lately, it feels like people, specifically managers who do not have engineering experience, are pushing the adoption of various AI tools. Honestly, it feels like an unthinking melee. I consider myself a conscientious objector to the use of AI, especially generative AI; I'm not fully opposed to it, but I constantly ask who actually benefits from the application of AI and what its financial, human, and environmental costs are beyond what is right in front of our noses. Yet, as a rank-and-file employee, I find myself with no real avenue to relay those concerns to people who have actual power to decide. Worse, I feel that even voicing such concerns, admittedly running against the almost blind optimism that I assume affects most marketing companies, is turning me into a pariah in my own workplace.

Continued

So my question is this: Is it worth it trying to encourage critical AI use in my company, or should I tone it down if only to keep paying the bills?

What would be your recommendation?

- Fight for your ideas?
- Quit from the company?
- Tone it down?
- ...

Definition (Whistle-blowing)

Disclosure of abuses in a company by an employee, without the consent of his superiors

Active Responsibility: Whistle Blowing

Definition (Whistle-blowing)

Disclosure of abuses in a company by an employee, without the consent of his superiors

- Ex-NSA Ex-CIA computer security expert, **Edward Snowden**, leaked highly classified information showing that CIA as doing massive illegal espionage.
- Ex-HSBC system engineer, **Herve Falciani**, leaked tax evaders data showing the extend of tax evasion and how the bank was supporting it
- Ex-Facebook product manager, **Francis Haugen** leaked internal documents showing that Meta knows that social networks may be harmful

Whistle-blowers: Heros or Villains?

- Violates deontological rules (e.g. keep confidentiality)
- Violates professional values (e.g. loyalty, trustworthiness)

It is an unsatisfactory way to deal with the tension between engineers and managers. It causes big sacrifices.

When is whistle-blowing morally required ? (i.e, not doing it would have been morally wrong)

- The company does serious **harm** to the public (customers or bystanders)
- The would-be whistle-blower has identified the harm, **reported** it to her superiors, and concluded that they will do nothing
- The would-be whistle-blower has **exhausted other internal procedures** within the company
- The would-be whistle-blower has **evidence** that would convince a reasonable, impartial observer
- The would-be whistle-blower has good reason to believe that revealing the information **will prevent or decrease** the harm

Study Case

Maria works in a tech company and her immediate boss, Martha, traveled to a professional conference in Hawaii. Because it was a professional trip, it was paid by the company. By chance, Maria finds out that Martha didn't attend any session of the conference. Instead, she attended a snorkeling course.

Do you think she **MUST** blow the whistle and **tell everybody** what she knows?

Study Case

Maria works in a tech company and her immediate boss, Martha, traveled to a professional conference in Hawaii. Because it was a professional trip, it was paid by the company. By chance, Maria finds out that Martha didn't attend any session of the conference. Instead, she attended a snorkeling course.

Do you think she **MUST** blow the whistle and **tell everybody** what she knows?

Solution: to make an objective judgment we need to confront the story with the four conditions.

Study Case

- Unless the company is in a very difficult economic situation, we cannot say that what Martha seriously harmed anybody. Therefore, this condition does not hold.
- From what we know, it seems that Maria did not report what she knows using the internal channels that the company may provide. Before going public, she should try internal channels keeping confidentiality and avoiding the spread of the information. Therefore, this condition does not hold.
- From the story it seems that Maria has certainty about what she knows. Therefore, this condition holds.
- It is not clear that making this information public that type of behavior will disappear. Therefore, I think this condition does not apply. But this is just my personal opinion.

Since in this case several conditions do not seem to hold, we can conclude that **Maria has no moral obligation to blow the whistle**. However, note that this does not mean that blowing the whistle would be morally wrong.

Study Case: the right to warn letter

The Right to Warn Letter

In June 2024, 13 current and former employees of OpenAI and Google DeepMind wrote an open letter demanding that the companies grant them a right to warn about the risks of advanced artificial intelligence.

Open Letter

Study Case: the right to warn letter

- That **the company will not discourage criticism** of the company for risk-related concerns, nor retaliate for risk-related criticism
- That **the company will facilitate** a verifiably anonymous process for current and former employees **to raise risk-related concerns to the companys board**, to regulators, and to an appropriate independent organization with relevant expertise;
- That **the company will support a culture of open criticism** and allow its current and former employees to raise risk-related concerns about its technologies (...), so long as trade secrets are appropriately protected;
- That **the company will not retaliate** against current and former employees who **publicly** share risk-related confidential information **after other processes have failed**.

Study Case: the right to warn letter

...

We accept that any effort to report risk-related concerns should avoid releasing confidential information unnecessarily. Therefore, once an adequate process for anonymously raising concerns to the company's board, to regulators, and to an appropriate independent organization with relevant expertise exists, we accept that concerns should be raised through such a process initially. **However, as long as such a process does not exist, current and former employees should retain their freedom to report their concerns to the public.**

EU Whistleblowing Directive (16/12/2019)

- Goal: To detect and prevent misconduct
- Establish effective, confidential and secure reporting channels
- Protect and enable whistleblowers without fear of retaliation.

[More info here](#)

Codes of Conduct (deontological code)

Codes of Conduct

Guidelines given by organizations for responsible behaviour of their members

- Institutions should value and expect high ethical *standards* for their members.
- Institutions must demonstrate their *true concern* regarding the ethical norms within the organization.
- Institutions should provide ethical *training* for their members.

It helps individuals to **feel protected** when behaving ethically

Two types of codes are relevant to engineers:

- **Professional** Codes (formulated by professional associations)
 - Integrity, honesty, competence,...
 - Obligation towards clients and employers (loyalty, confidentiality,...)
 - Social Responsibility (safety, welfare, environmental,...)
- **Corporate** Codes (formulated by companies)
 - Mission statement
 - Core Values (e.g. teamwork, responsibility,...)
 - Responsibility to stakeholders
 - Norms and Rules (e.g. acceptance of gifts, sentimental relationships within work,...)

Example: Corporate Code of Conduct

Airbus

Example: Professional Code of conduct

ACM Code of Ethics and Professional Conduct ([link here](#)): **4 Sections:**

- 1 Fundamental ethical considerations
- 2 Specific ethical considerations
- 3 For people in leadership roles
- 4 Principles involving compliance with the code

ACM Code of Ethics (Section 1)

- **Contribute** to society and human well being
 - dark pattern interfaces (also known as a "deceptive design pattern")?
 - Scheduled Obsolescence (Apple and Samsung 2018)?
- **Avoid harm** to others
 - Volkswagen pollution measures?
- **Be honest** and trustworthy (Do not make false claims about the system)
 - Security (e.g. critical systems, money transfer)
 - Privacy (e.g. Grindr)
 - Quality (e.g. optimality)

ACM Code of Ethics (Section 1)

- **Be fair** and take action not to discriminate
- Honor **property rights**
 - Protect **copyrights**, **patents** and **secrets** owned by your company
 - Give proper credit for intellectual property
- Respect **privacy**
 - Take precautions to prevent unauthorized access even accidental disclosures
 - Avoid data snooping (e.g. programmers have access to databases)

Case Study: Dark Patterns

Definition (Dark Pattern)

also known as a "deceptive design pattern", is a user interface that has been carefully crafted to trick users into doing things.

Example

Dark Patterns violate many Principles of the ACM code of Ethics. They can make users feel fooled (Principle 1.2), provide deliberately misleading information (Principle 1.3), or discriminate against those with disabilities (Principle 1.4). Computing professionals have a moral obligation to use their skills to benefit the members of society (Principle 1.1), not to deceive them. Furthermore, the use of dark patterns is an affront to the dignity of users, violating Principle 2.1.

Example (San Bernardino Shooting)

- On 2015, 14 people were killed in a terrorist attack in San Bernardino, California.
- Apple employees refused to perform the software engineering tasks that would be necessary to provide the F.B.I. with access to the contents of an iPhone used by one of the shooters. They justified their behaviour on the ACM code.
- ACM urges computer professionals to obey existing laws unless they raise ethical issues that come into play around **privacy** and technological systems that have the ability to harm the public

The Ethical Cycle

Ethical cycle (van de Poel, Royakkers, 2007)

A methodological tool to **structure our analysis** on how to deal with Professional Ethical Dilemmas.

Iterative process (follow steps, but backtrack when needed)

- ➊ Moral Problem Statement (precise statement)
- ➋ Problem Analysis (gather context and relevant data)
- ➌ Options for Action (identify alternatives)
- ➍ Ethical Evaluation (analyze alternatives)
- ➎ Reflection (compare alternatives)
- ➏ Moral acceptable action (chose alternative)

Iterative process (follow steps, but backtrack when needed)

- ① Moral Problem Statement
 - What is the problem?
 - Who has to decide?
 - The moral nature of the problem (values and/or norms endangered)
- ② Problem Analysis
- ③ Options for Action
- ④ Ethical Evaluation
- ⑤ Reflection
- ⑥ Moral acceptable action

Iterative process (follow steps, but backtrack when needed)

- ➊ Moral Problem Statement
- ➋ Problem Analysis
 - Stakeholders (agents affected, consequences)
 - Relevant facts (uncertainty,...)
 - Moral values (conflicting values)
 - Norms (tensioned deontological rules)
- ➌ Options for Action
- ➍ Ethical Evaluation
- ➎ Reflection
- ➏ Moral acceptable action

Iterative process (follow steps, but backtrack when needed)

- ① Moral Problem Statement
- ② Problem Analysis
- ③ Options for Action
 - Obvious alternatives
 - Creative alternatives (e.g. tradeoffs,...)
 - Cooperation strategy (look for diff. points of view)
- ④ Ethical Evaluation
- ⑤ Reflection
- ⑥ Moral acceptable action

Iterative process (follow steps, but backtrack when needed)

- ① Moral Problem Statement
- ② Problem Analysis
- ③ Options for Action
- ④ Ethical Evaluation
 - Arg. by analogy (think of how analogous cases were solved)
 - Utilitarianism (consequences on values)
 - Kant (norms, professional codes, corporate codes)
 - Virtue Ethics (professional virtues)
- ⑤ Reflection
- ⑥ Morally acceptable action

Iterative process (follow steps, but backtrack when needed)

- ➊ Moral Problem Statement
- ➋ Problem Analysis
- ➌ Options for Action
- ➍ Ethical Evaluation
- ➎ Reflection
 - Think now how do you feel with respect to the different options
- ➏ Moral acceptable action

The Ethical Cycle: Example

A civil engineer working for a local government has \$500000 to spend on road safety. After some consideration, she narrows her choice to two safety improvements (same cost) on two possible intersections where (deadly) accidents occur. Intersection A (urban area) and intersection B (rural area).

The Ethical Cycle: Example

Problem Statement:

- What is the problem?
- Who has to act? (is a civil engineer the right person to make the decision? may be an urbanist, politician?)
- Is it an ethical problem?
 - **values:** safety, fairness
 - **norms:** policy makers should spend taxes' money efficiently

Variations where the dilemma is not ethical:

- where to build a swimming pool

The Ethical Cycle: Example

Problem analysis:

- **Stakeholders:** Drivers and their passengers, pedestrians (neighbors), tax payers,...

The Ethical Cycle: Example

Problem analysis:

- **Stakeholders:** Drivers and their passengers, pedestrians (neighbors), tax payers,...
- **Facts A:** 20000 vehicles/day, 2 fatalities/year, 6 injuries/year, 40000 property damage/year
- **Facts B:** 5000 vehicles/day, 1 fatalities/year, 2 injuries/year, 12000 property damage/year

The Ethical Cycle: Example

Problem analysis:

- **Stakeholders:** Drivers and their passengers, pedestrians (neighbors), tax payers,...
- **Facts A:** 20000 vehicles/day, 2 fatalities/year, 6 injuries/year, 40000 property damage/year
- **Facts B:** 5000 vehicles/day, 1 fatalities/year, 2 injuries/year, 12000 property damage/year
- **Exp.Impr. A:** reduction fatalities 50%, reduction injuries 50%, reduction damage 25%
- **Exp.Impr. B:** reduction fatalities 50%, reduction injuries 50%, reduction damage -25% (increment)

The Ethical Cycle: Example

Problem analysis:

- In this problem the **uncertainty** is very clear
- Note that A is safer than B (Risk of death factor $2/20000$ vs $1/5000$)
- also in terms of injuries ($6/20000$ vs $2/5000$) and damage ($4/2$ vs $12/5$)
- But A is more deadly than B

The Ethical Cycle: Example

Options for Action

- A
- B
- Do nothing
- Creative alternatives/cooperation: may be by talking to neighbors, near cities, ... Investigate about other sources of money, do one option now and allocate future money for the other,...

Ethical Evaluation:

- **Common sense/Virtue Ethics:**

- Saving lives is what really matters here (other damages are less important)
- "Do nothing" is out
- improving A saves more lives (1 vs 0.5 per year)
- improving B makes city fairer in terms of safety

The Ethical Cycle: Example

Ethical Evaluation:

- **Utilitarian:**

- **weighting schemes:** fatality ($\$10^6$, see "value of life" in wikipedia), injury ($\$30000$), damage
- Value of A: $10^6 + 90000 + 40000 = 1130000$
- Value of B: $0.5 \times 10^6 + 30000 + 12000 + 3000 = 545000$
- Note that the value of A and B is "per year" and the cost of it is 500000

Site A has the largest benefit (even assuming errors in uncertainty and weighting schemes)

The Ethical Cycle: Example

Ethical Evaluation:

- **Kantian:**

- A fundamental principle of democracy is that all people are equally important.
- Everybody has the right to the same level of protection (in a **fair** society)

Site B improves fairness

The Ethical Cycle: Example

Reflection

- People living in rural areas must assume that they will not be treated fairly in terms of infrastructures
- Safety infrastructures are more important than, say, a gym.
- The current risk factor is double in B
- Choosing option A, it will quadruple it
- Choosing option B, it will equal it

Moral acceptable action

I would choose B, not so much because it equals the risk factor, but because choosing A would make A four times safer than B (and that makes me uncomfortable)

Historically, technology is developed by companies whose main incentive is profit. Although this model has proven very effective, sometimes its not optimum in terms of social impact

Historically, technology is developed by companies whose main incentive is profit. Although this model has proven very effective, sometimes its not optimum in terms of social impact

Humane Technology

Design and Develop Technology in a way that supports our well-being, democratic functioning, and shared information environment.

- That is: Take Ethics into account.

We are going to see two approaches to that.

The Three Laws of Humane Tech

- Proposed by **Center for Humane Technology**
- **Purpose:** Simplify what does it mean to be a responsible technologist
- **Motivation:** Aza Raskin invented the infinite scroll in 2006
 - Invented in one context (yelp)
 - Adopted by social media (their business is to capture users attention)
 - **Effect:** 200,000 human lifetimes per day being *wasted*
 - **His mindset in 2007:** infinite scroll is cool, so let's do it
 - **His mindset now:** sorry world! I did not think of the consequences of what I was doing.

The Three Laws of Humane Tech

The laws

- ① When you invent a new technology, you uncover a new class of responsibilities. And it's not always obvious what those responsibilities are.
- ② If that invention, confers power, it will start a race (among companies).
- ③ If you do not coordinate and set norms, that race will end in tragedy.

The Three Laws of Humane Tech

Examples:

- Cheap digital cameras → right to privacy
- Cheap storage → right to be forgotten
- Retweet button → right to faithful information
- Notifications → right to own your attention

Engineers Active Responsibility

- Be aware of these laws
- Anticipate ethical issues before it is too late
- Stand up for coordination and setting norms

Ethics in Design

Engineers design products. Design decisions have social consequences that need to be unveiled and analyzed. Ethics must be considered.

Ethics in Design

Engineers design products. Design decisions have social consequences that need to be unveiled and analyzed. Ethics must be considered.

Example:

- Car safety is tested with dummies based on man-like bodies
- individual dummies can each cost more than \$1 million
- women are more likely to suffer injuries in car crashes than men
- Calls for an accurate female crash test dummy date back decades

- Cars are not well designed for women (same for building halls, some drugs,...)
- Buildings are not well designed for handicapped
- Electronic payment is not well designed for homeless
- ...

Humane Design, <https://humanebydesign.com/>

designing ethically humane digital products through patterns focused on user well-being

Moral Design

The deliberate design of a product aiming at users acting ethically

- Speed bumps
- Metro tourniquets
- A counterexample: Rober Moses' racist Overpasses

Sometimes, simple design ideas may have big impact

- Seatbelt warning signals
- Hide "share" button in social media

Humane Design: Summary

- ① When there is a new technology: stand for precaution and coordination (three laws of Humane Tech)
- ② When designing a new product, be socially responsible:
 - Take into account and be fair with all the affected stakeholders (design social responsibility)
 - Consider adding features that guarantee a good use of the technology (design morality)

AI has become one of the most influential new technologies. Researchers have warned about its risks and have demanded regulation.

AI has become one of the most influential new technologies. Researchers have warned about its risks and have demanded regulation.

Most popular current AI systems

- Generate **plausible** content (sounds very reasonable)
- They **never say I do not know the answer** (because they do not have the notion of right and wrong, they are *stochastic systems*)

AI has become one of the most influential new technologies. Researchers have warned about its risks and have demanded regulation.

Most popular current AI systems

- Generate **plausible** content (sounds very reasonable)
- They **never say I do not know the answer** (because they do not have the notion of right and wrong, they are *stochastic systems*)

That is very powerful for problems such as,

- There is a lot of data
- The notion of correctness,
 - is not very relevant or,
 - it can be easily checked

But it is dangerous if used in other contexts

Is AI good or bad?

Dan Brown

Humans have never created any technology that have later transformed into a weapon

Is AI good or bad?

Dan Brown

Humans have never created any technology that have later transformed into a weapon

- As any other technology IA is **good**, **bad** and **unavoidable**
- It is our human duty to make a responsible use of AI

What is the main risk of AI: Human stupidity

Cognitive Vulnerabilities:

- We do not like to make hard decisions
- We take decisions out of intuition and later we look for consistent argumentation justifying it.
- We love to be told that we are right
- We anthropomorphize too quickly

What is the main risk of AI: Human stupidity

Cognitive Vulnerabilities:

- We do not like to make hard decisions
- We take decisions out of intuition and later we look for consistent argumentation justifying it.
- We love to be told that we are right
- We anthropomorphize too quickly

AI risks due to human vulnerability

- AI is very **persuasive** (designed to convince us)
- AI is **sycophantic** (designed to please us).

What is the main risk of AI: Human stupidity

Cognitive Vulnerabilities:

- We do not like to make hard decisions
- We take decisions out of intuition and later we look for consistent argumentation justifying it.
- We love to be told that we are right
- We anthropomorphize too quickly

AI risks due to human vulnerability

- AI is very **persuasive** (designed to convince us)
- AI is **sycophantic** (designed to please us).
- If we **anthropomorphize**, we may give to it unreasonable licenses affecting our live.

What is the main risk of AI: Human stupidity

The New York Times

<https://www.nytimes.com/2025/11/23/technology/openai-chatgpt-users-risks.html>

What OpenAI Did When ChatGPT Users Lost Touch With Reality

In tweaking its chatbot to appeal to more people, OpenAI made it riskier for some of them. Now the company has made its chatbot safer. Will that undermine its quest for growth?



Listen to this article · 23:14 min [Learn more](#)

Risk Example: Behaviour Manipulation

- It is known that **humans are easily manipulable**
 - **Marketing** has been always exploiting it
- Current technology makes it much more effective:
 - **More info:** we give detailed sensitive data
 - **More exposure:** we are all the time exposed

Cambridge Analytica Scandal

During the Brexit referendum, they identified **neurotic** people and exposed them with **fear-based** political ads

Risk Example: Unfair Classification

Algorithmic Fairness

If two individuals are identical except for a prejudice variable (gender, race, religion, sexual orientation,...) they should be treated identically by algorithms

There have been observed several cases of unfairness (www.propublica.org):

- Racial prejudices by crime forecasters
- Racial prejudices by real state advertisement
- Gender prejudices by job opening advertisement

Algorithmic fairness

Algorithms can be unfair because:

- The developers are biased
- The training data may be biased

Fairness and Reinforcement Learning

Artifacts (including algorithms) are designed to work well for the majority. Sometimes this causes an unfair reinforcement process.

Consider the design of a hiring mechanism.

- If it is tuned with an initial mostly masculine population it will take into account masculine features correlated to success.
- The few women selected may be selected for the wrong reason, so they may not be the best women candidates
- If the algorithm is updated it will be biased against women

Fairness and Reinforcement Learning

Artifacts (including algorithms) are designed to work well for the majority. Sometimes this causes an unfair reinforcement process.

Consider the design of a hiring mechanism.

- If it is tuned with an initial mostly masculine population it will take into account masculine features correlated to success.
- The few women selected may be selected for the wrong reason, so they may not be the best women candidates
- If the algorithm is updated it will be biased against women

The Guardian 2018

Amazon ditched AI recruiting tool that favored men for technical jobs

Risk Example: Data Recording and Privacy

Many platforms collect data for profit. This information can have a huge impact on you.

Limits on Recording and use

What is ethical depends on each case:

- Some data should not be collected (apps should collect only what they need)
- Some data should be recorded but not used (surveillance)
- Some data should be recorded and only used to its purpose (phone companies must know your location)

Limits on Recording and use

But we have to take into account:

- Sometimes critical data can be *inferred* by crossing data
 - We are not aware of the real magnitude of that (e.g. 20 likes from facebook)
- Sometimes we are willing to give away our data in exchange of free-service.
 - We are not aware of the real magnitude of that (e.g. info about our house from Romba or Instagram)
 - We are not aware of the value of our data

A warning on privacy and data anonymization

Anonymization

Removal of identifiable information from data.

- Given Zip code, date of birth and sex, a person can be identified 87% of times.
- In 2006 Netflix published 10 million anonymous movie rankings by 500,000 customers. The anonymous Netflix records crossed with crowd-sourced critic sites re-identified many users.

A warning on privacy and data anonymization

"The suspected Golden State Killer was caught 40 years later because his relative's DNA was available on a genealogy website"



Shared data

That story had a happy ending, but it shows that when you give your DNA to someone, you are partially giving the DNA of your close relatives. That raises the question of who is the owner of your DNA?

First legal framework for AI systems in the EU Seven Principles:

- human agency and oversight;
- technical robustness and safety;
- privacy and data governance;
- transparency;
- diversity, non-discrimination and fairness;
- societal and environmental well-being
- accountability.

- Classifies systems by **risk to users**:
 - **Unacceptable Risk**: forbidden
 - **High Risk**: obligations w.r.t. learning datasets, data management, cybersecurity, human-in-the-loop
 - **Low Risk**: obligations on transparency
 - **Minimum Risk**

Responsability

Mainly on developers and providers, but also on users

Cost?

Some people warned that it would slow down progress and put Europe in disadvantage w.r.t. US and China.

Classification of AI systems

Definition (Unacceptable)

A very limited set of uses of AI that are particularly **harmful** and contravene EU values, as they **infringe fundamental rights** and are therefore **prohibited**.

Classification of AI systems

Definition (Unacceptable)

A very limited set of uses of AI that are particularly **harmful** and contravene EU values, as they **infringe fundamental rights** and are therefore **prohibited**.

- Manipulative techniques that alter behaviour causing significative harm

Classification of AI systems

Definition (Unacceptable)

A very limited set of uses of AI that are particularly **harmful** and contravene EU values, as they **infringe fundamental rights** and are therefore **prohibited**.

- Manipulative techniques that alter behaviour causing significant harm
- Exploit vulnerabilities altering behaviour causing significant harm

Classification of AI systems

Definition (Unacceptable)

A very limited set of uses of AI that are particularly **harmful** and contravene EU values, as they **infringe fundamental rights** and are therefore **prohibited**.

- Manipulative techniques that alter behaviour causing significant harm
- Exploit vulnerabilities altering behaviour causing significant harm
- Classify individuals on the basis of social or personal (known or inferred) behaviour for detrimental treatment

Classification of AI systems

Definition (Unacceptable)

A very limited set of uses of AI that are particularly **harmful** and contravene EU values, as they **infringe fundamental rights** and are therefore **prohibited**.

- Manipulative techniques that alter behaviour causing significant harm
- Exploit vulnerabilities altering behaviour causing significant harm
- Classify individuals on the basis of social or personal (known or inferred) behaviour for detrimental treatment
- Risk assessment for crime prediction based only on profiling

Classification of AI systems

Definition (Unacceptable)

A very limited set of uses of AI that are particularly **harmful** and contravene EU values, as they **infringe fundamental rights** and are therefore **prohibited**.

- Manipulative techniques that alter behaviour causing significant harm
- Exploit vulnerabilities altering behaviour causing significant harm
- Classify individuals on the basis of social or personal (known or inferred) behaviour for detrimental treatment
- Risk assessment for crime prediction based only on profiling
- Facial recognition databases (massively)

Classification of AI systems

Definition (Unacceptable)

A very limited set of uses of AI that are particularly **harmful** and contravene EU values, as they **infringe fundamental rights** and are therefore **prohibited**.

- Manipulative techniques that alter behaviour causing significant harm
- Exploit vulnerabilities altering behaviour causing significant harm
- Classify individuals on the basis of social or personal (known or inferred) behaviour for detrimental treatment
- Risk assessment for crime prediction based only on profiling
- Facial recognition databases (massively)
- Manipulate emotions in workplace or educational institutions

Classification of AI systems

Definition (Unacceptable)

A very limited set of uses of AI that are particularly **harmful** and contravene EU values, as they **infringe fundamental rights** and are therefore **prohibited**.

- Manipulative techniques that alter behaviour causing significant harm
- Exploit vulnerabilities altering behaviour causing significant harm
- Classify individuals on the basis of social or personal (known or inferred) behaviour for detrimental treatment
- Risk assessment for crime prediction based only on profiling
- Facial recognition databases (massively)
- Manipulate emotions in workplace or educational institutions
- Biometric categorization that induce race, religion, political or sexual orientation

Classification of AI systems

Definition (Unacceptable)

A very limited set of uses of AI that are particularly **harmful** and contravene EU values, as they **infringe fundamental rights** and are therefore **prohibited**.

- Manipulative techniques that alter behaviour causing significant harm
- Exploit vulnerabilities altering behaviour causing significant harm
- Classify individuals on the basis of social or personal (known or inferred) behaviour for detrimental treatment
- Risk assessment for crime prediction based only on profiling
- Facial recognition databases (massively)
- Manipulate emotions in workplace or educational institutions
- Biometric categorization that induce race, religion, political or sexual orientation
- Real-time remote biometric identification in public spaces for law enforcement

Classification of AI systems

Definition (High Risk)

limited number of AI systems that may have an **adverse impact** on the **security** of individuals or their **fundamental rights** under the EU Charter of Fundamental Rights

Classification of AI systems

Definition (High Risk)

limited number of AI systems that may have an **adverse impact** on the **security** of individuals or their **fundamental rights** under the EU Charter of Fundamental Rights

- Systems used to investigate, interpret legal rules, apply the law

Classification of AI systems

Definition (High Risk)

limited number of AI systems that may have an **adverse impact** on the **security** of individuals or their **fundamental rights** under the EU Charter of Fundamental Rights

- Systems used to investigate, interpret legal rules, apply the law
- Systems used to manage critical infrastructures

Classification of AI systems

Definition (High Risk)

limited number of AI systems that may have an **adverse impact** on the **security** of individuals or their **fundamental rights** under the EU Charter of Fundamental Rights

- Systems used to investigate, interpret legal rules, apply the law
- Systems used to manage critical infrastructures
- Systems used in education for admission, assessment, detect prohibited behaviour

Classification of AI systems

Definition (High Risk)

limited number of AI systems that may have an **adverse impact** on the **security** of individuals or their **fundamental rights** under the EU Charter of Fundamental Rights

- Systems used to investigate, interpret legal rules, apply the law
- Systems used to manage critical infrastructures
- Systems used in education for admission, assessment, detect prohibited behaviour
- Systems used in employee management (recruitment, performance appraisal)

Classification of AI systems

Definition (High Risk)

limited number of AI systems that may have an **adverse impact** on the **security** of individuals or their **fundamental rights** under the EU Charter of Fundamental Rights

- Systems used to investigate, interpret legal rules, apply the law
- Systems used to manage critical infrastructures
- Systems used in education for admission, assessment, detect prohibited behaviour
- Systems used in employee management (recruitment, performance appraisal)
- Systems used in essential services (e.g. determine recipients of social services)
- ...

Definition (Limited Risk)

risks associated with a **lack of transparency** in the use of AI and therefore the need for specific transparency obligations to ensure that humans are informed when necessary, building trust.

Definition (Limited Risk)

risks associated with a **lack of transparency** in the use of AI and therefore the need for specific transparency obligations to ensure that humans are informed when necessary, building trust.

- Content creation (e.g. video, text, sound,...)
- Interaction with humans (e.g. bots)
- General purpose models (e.g. Gemini)

Definition (Low Risk)

AI systems that can be developed and used in accordance with existing legislation without additional legal obligations.

Principles, Indicators and Observables

- <https://oeiac.cat/en/>
- Proposed by the Observatory for Ethics in Artificial Intelligence of Catalonia
- An assessment tool that facilitates compliance with current rules and regulations on risks associated with artificial intelligence through a comprehensive verification process.
- It contains more than 100 practical examples